PURNA SHAH

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SUMMARY

Computer Science student with a strong foundation in Python, SQL, and statistics. I have hands-on experience in data preprocessing, database management, and data analysis using Pandas and Excel. Seeking internship opportunities to apply technical skills in real-world projects. I enjoy hunting for the "why" hidden behind every data point.

EDUCATION

Rajarambapu Institute of Technology — B.Tech, Computer Science and Engineering — Aug 2023 – May 2027

V.R.D. Junior College of Science (H.S.C.) — Jun 2022 – Mar 2023

K.K. Gulavani Higher Secondary School (S.S.C.) — Jun 2020 – Mar 2021

TECHNICAL SKILLS

Languages: Python, C

Data Analysis: Pandas, NumPy, Matplotlib, Excel (Basic), Statistics

Core CS: Data Structures & Algorithms

Tools: Git, GitHub, Visual Studio Code

Other: Teamwork, Presentation, Event Coordination

PROJECTS

Multi-Modal ML Framework for Pancreatic Cancer Risk Stratification — Team project(ongoing)

Led development of multi-modal ML pipeline combining clinical and wearable physiological data for early cancer risk prediction.

- Engineered a Python preprocessing pipeline (missing-value imputation, normalization, categorical encoding, feature engineering) to prepare clinical datasets for modeling.
- Train and benchmarked ensemble classifiers (Random Forest, XGBoost) across 5 evaluation metrics — Accuracy, Precision, Recall, F1-score, and ROC-AUC — to select the top-performing model.
- Conducted EDA and feature-importance analysis to surface key clinical risk indicators, communicated through Matplotlib visualizations.
- Collaborated with the embedded systems team to integrate wearable sensor data into the pipeline, extending the model to multi-modal inputs.

Customer Churn Analysis (Telco Industry) — MySQL | Python | Pandas | SciPy | Excel

Independent analysis of a public IBM dataset covering 7,000+ telecom customers to uncover churn drivers.

- Designed a MySQL database and loaded 7,000+ customer records; wrote SQL queries to segment churn rate by contract type, tenure, and payment method.
- Applied chi-square hypothesis testing in Python ($p < 0.001$) to confirm contract type and payment method as statistically significant churn drivers.
- Created Python visualizations and an Excel summary workbook (pivot tables and charts) to present retention recommendations.

Cultural Event Management System — Java (Swing) | MySQL | JDBC

Desktop application for managing college cultural events, built end-to-end from database design to UI.

- Designed a 6-table MySQL schema and implemented a DAO layer using PreparedStatements for secure, parameterized CRUD operations.
- Implemented a Java Swing interface backed by JDBC for persistent data storage and retrieval, supporting real event and participant management workflows.

EXPERIENCE & LEADERSHIP

CodSoft Internship — Python Programming — Jun 2025 – Jul 2025

- Developed 3 Python applications (To-Do List, Calculator, Rock-Paper-Scissors) applying OOP, file handling, and the random module.
- Strengthened core programming fundamentals through hands-on implementation of functions and object-oriented design.

Non-Technical Head — Oyster Kode Club, RIT — 2023 – Present

- Coordinated 5+ college-wide events and competitions for 80+ attendees.
- Ran weekly activities to build communication and teamwork skills among junior club members.

CERTIFICATIONS

- NPTEL — Database Management System
- GeeksforGeeks — Python (Basics)

LANGUAGES

English (Professional Working) • Hindi (Native/Bilingual) • Marathi (Native/Bilingual) • Gujarati (Professional Working) • Spanish (Elementary)